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Topical Red Palm Oil Modulates Cutaneous Oxidative-Stress Biomarkers in Streptozotocin-Induced Diabetic Wistar Rats

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ABSTRACT:

Diabetes-associated oxidative stress contributes to cutaneous injury and may weaken local antioxidant defense. Red palm oil (RPO), a natural oil rich in carotenoids and vitamin E isomers, has potential as a topical antioxidant, but its direct biochemical effect on diabetic skin remains insufficiently defined. This study evaluated the effect of 10% topical RPO on superoxide dismutase (SOD) activity and malondialdehyde (MDA) concentration in streptozotocin-induced diabetic Wistar rat skin. A controlled post-test-only laboratory experiment was performed using 27 male Wistar rats allocated into three groups: non-diabetic control (ND), diabetic control (DM), and diabetic rats treated with 10% RPO (DM+RPO), with nine rats in each group. Diabetes was induced by a single intraperitoneal injection of streptozotocin at 50 mg/kg body weight and confirmed by blood glucose >200 mg/dL. RPO or base cream was applied to a 3 x 3 cm dorsal area twice daily for 28 days. On day 29, full-thickness dorsal skin was collected for colorimetric SOD and MDA analysis. SOD was analyzed using one-way ANOVA followed by Tukey testing, whereas MDA was analyzed using Kruskal-Wallis and Holm-adjusted pairwise Mann-Whitney U testing. SOD activity differed significantly among groups ($F=23.696$, $p<0.001$), increasing from 32.92 ± 4.33 U/mL in DM rats to 39.88 ± 2.13 U/mL in DM+RPO rats. MDA also differed significantly ($H=16.358$, $p<0.001$), decreasing from 10.24 ± 2.00 to 7.59 ± 2.02 nmol/mL after RPO treatment. SOD and MDA were negatively correlated ($r=-0.544$, $p=0.003$). Topical RPO improved the cutaneous antioxidant-oxidant biomarker profile in diabetic rats and deserves further formulation, safety, and mechanistic evaluation.

KEYWORDS: Red palm oil, Superoxide dismutase, Malondialdehyde, Diabetic skin, Wistar rats.

INTRODUCTION:

Diabetes mellitus is a major global health problem with consequences extending beyond glycemic dysregulation. The International Diabetes Federation reported that 589 million adults aged 20-79 years were living with diabetes in 2024 and projected an increase to 853 million by 2050. Indonesia was also listed among countries with a high diabetes burden, with more than 20 million affected adults in 2024. ¹ These epidemiological data are relevant to skin-related diabetic complications because chronic hyperglycemia contributes to microvascular dysfunction, impaired tissue integrity, and susceptibility to ulcers and infection. Diabetic foot ulcer risk has been reported to be substantial over a patient's lifetime and is associated with hospitalization, amputation, and mortality. ²

Oxidative stress is a central biological process linking diabetes to peripheral tissue injury. Hyperglycemia and mitochondrial dysfunction can increase production of reactive oxygen species (ROS), while persistent ROS exposure may damage cell membranes, disrupt repair responses, reduce proliferative capacity, and prolong inflammation. ^{3,4} In the skin, antioxidant enzymes are important for maintaining redox balance. Superoxide dismutase (SOD) neutralizes superoxide radicals and is commonly used as an indicator of antioxidant defense, whereas malondialdehyde (MDA) is a lipid peroxidation product that reflects oxidative membrane injury. ⁵ Therefore, the combined assessment of SOD and MDA provides a useful biochemical approach for evaluating antioxidant-oxidant balance in diabetic skin.

Topical antioxidant strategies have increasingly been explored because local application may deliver bioactive agents to affected skin while minimizing systemic exposure. In diabetic wound or skin-related models, chrysin and topical metformin have been shown to improve antioxidant status and reduce oxidative-damage markers. ^{6,7} Topical *Phaleria macrocarpa* extract has also been reported to support inflammatory and proliferative phases in diabetic rat wounds. ⁸ These findings indicate that topical natural products and antioxidant preparations can modulate biochemical and cellular processes relevant to diabetic tissue injury, although the biological response depends on the compound, vehicle, and experimental model used.

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Red palm oil (RPO), derived from *Elaeis guineensis*, is of pharmaceutical interest because it contains lipophilic constituents such as carotenoids, tocopherols, tocotrienols, polyphenols, phytosterols, squalene, and coenzyme Q10.^{9,10} These compounds are relevant to topical delivery because the stratum corneum is lipid-rich, and lipophilic antioxidants may interact with cutaneous lipid compartments. In addition, the retention of beta-carotene and vitamin E during RPO processing supports its potential use as a natural antioxidant source.⁹ The health and cosmetic relevance of RPO bioactive constituents has also been discussed in recent literature.¹⁰

Previous animal studies have mainly evaluated RPO through systemic or non-cutaneous models. RPO has been reported to ameliorate oxidative challenge and inflammatory responses through pathways involving NF-kappaB and Nrf2/GCL/HO-1 signaling, and pretreatment with RPO increased antioxidant enzyme activity while reducing MDA in Wistar rats exposed to oxidative toxic injury.^{11,12} In the dermatological field, palm tocotrienol-rich fractions, topical tocotrienol preparations, nanoemulsified tocotrienol-rich systems, and vitamin E derived from RPO have shown biological activities relevant to cutaneous repair.¹³⁻¹⁶ However, these studies do not fully answer whether topical RPO as an intact natural-oil-based preparation can directly improve SOD and MDA balance in diabetic skin without a wound model.

The research gap is therefore specific: although topical antioxidants and palm-derived bioactives have been investigated in related models, direct evidence on topical RPO and cutaneous SOD-MDA biomarkers in diabetes remains limited. The present study was designed as a controlled preclinical experiment to evaluate whether topical 10% RPO modulates SOD activity and MDA concentration in the dorsal skin of streptozotocin-induced diabetic Wistar rats. The study was positioned as a biochemical investigation of diabetic skin oxidative stress rather than as a wound-healing efficacy study.

MATERIALS AND METHODS:

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Study design: A controlled laboratory experiment with a post-test-only control-group design was conducted. The study included three parallel groups: a non-diabetic control group (ND), a diabetic control group (DM), and a diabetic group treated with 10% topical RPO (DM+RPO). Each group consisted of nine rats, giving a total sample size of 27 animals.

The post-test-only approach was appropriate for this experiment because tissue biomarkers were assessed after completion of the intervention and tissue excision was required for biochemical analysis. The three-group structure allowed differentiation between the baseline non-diabetic condition, the effect of diabetes without active treatment, and the effect of topical RPO under diabetic conditions. This structure is useful for evaluating whether RPO shifts oxidative-stress biomarkers toward a non-diabetic pattern without assuming clinical efficacy.

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Animals and ethical approval: Wistar rats aged 8-12 weeks and weighing 150-200 g were housed under standard laboratory conditions with a light/dark cycle at 22±3°C and free access to food and water. Animals were acclimatized before the intervention. The study protocol was approved by the Health Research Ethics Committee, Universitas Sumatera Utara, Indonesia (No. 756/KEPK/USU/2024; approved 28 June 2024), and procedures were conducted in accordance with the approved institutional protocol for laboratory animals.

Diabetes induction and experimental groups: Rats were fasted for 12 hours before diabetes induction to standardize metabolic conditions. Diabetes was induced by a single intraperitoneal injection of streptozotocin (STZ) at 50 mg/kg body weight dissolved in 0.1 M citrate buffer at pH 4.5. Five days after injection, blood glucose was measured from the tail vein. Rats with blood glucose levels >200 mg/dL were classified as diabetic. The ND group did not receive STZ and received base cream. The DM group received STZ and base cream. The DM+RPO group received STZ and 10% topical RPO.

Topical RPO intervention: The RPO material used for the topical preparation was red palm stearin, the stearin fraction of red palm oil. Dorsal hair was carefully shaved before treatment to ensure consistent contact with the application area. The 10% RPO preparation was applied topically to a 3 x 3 cm area of dorsal skin twice daily for 28 consecutive days. The ND and DM groups received base cream at the same frequency and duration to maintain comparable handling and topical exposure across groups.

The dorsal application area was selected to provide a defined and reproducible skin surface for repeated topical exposure. Because the study evaluated biochemical changes in skin rather than systemic metabolic control, the topical route was used to focus on local cutaneous effects. The base-cream control was included to reduce bias from handling and vehicle exposure in the ND and DM groups.

Tissue collection and biochemical measurements: On day 29, rats were anesthetized in accordance with the approved laboratory protocol, and full-thickness dorsal skin tissue was excised from the treated area. Tissue samples were prepared for biochemical analysis of antioxidant defense and lipid peroxidation. SOD activity and MDA concentration

were determined using colorimetric methods and expressed as U/mL and nmol/mL, respectively. Tissue protein content was also measured using the Bradford method to support biochemical interpretation and comparability among samples. SOD and MDA were selected because they represent complementary components of the oxidative-stress profile. SOD reflects enzymatic antioxidant defense against superoxide radicals, whereas MDA reflects lipid peroxidation resulting from oxidative damage to membrane lipids. The use of both markers therefore allowed assessment of whether topical RPO influenced antioxidant capacity and oxidative injury in the same tissue context.

Statistical analysis: Normality was assessed using the Shapiro-Wilk test and variance homogeneity using Levene's test. SOD data met parametric assumptions and were analyzed using one-way analysis of variance (ANOVA), followed by Tukey post hoc testing. MDA data did not meet normality assumptions in all groups and were analyzed using the Kruskal-Wallis test, followed by pairwise Mann-Whitney U testing with Holm correction. The association between SOD activity and MDA concentration was assessed using Pearson correlation analysis based on the available correlation output. All tests were two-sided, and $p < 0.05$ was considered statistically significant.

RESULTS:

The analysis included all 27 rats, with nine animals in each experimental group. Table 1 presents descriptive values for cutaneous SOD activity and MDA concentration. The diabetic control group showed the lowest mean SOD activity (32.92 ± 4.33 U/mL), whereas SOD values were higher in the non-diabetic control group (42.35 ± 1.99 U/mL) and the RPO-treated diabetic group (39.88 ± 2.13 U/mL). Conversely, MDA concentration was highest in the diabetic control group (10.24 ± 2.00 nmol/mL), followed by the RPO-treated diabetic group (7.59 ± 2.02 nmol/mL), and lowest in the non-diabetic control group (5.45 ± 1.19 nmol/mL).

Relative to the ND group, diabetes reduced mean SOD activity by approximately 22.3% and increased mean MDA concentration by approximately 87.9%. Compared with the DM group, RPO treatment increased mean SOD activity by approximately 21.1% and reduced mean MDA concentration by approximately 25.9%. These descriptive changes support the direction of the statistical findings and show that RPO treatment partially reversed, but did not completely normalize, the diabetic oxidative-stress pattern.

Table 1. Cutaneous SOD activity and MDA concentration by experimental group.

Outcome	ND (n=9)	DM (n=9)	DM+RPO (n=9)
SOD activity (U/mL)	42.35 ± 1.99^b	32.92 ± 4.33^a	39.88 ± 2.13^b
MDA concentration (nmol/mL)	5.45 ± 1.19^a	10.24 ± 2.00^c	7.59 ± 2.02^b

Values are mean $\pm$ SD. Different superscript letters within the same row indicate significant pairwise differences ($p < 0.05$). ND, non-diabetic control; DM, diabetic control; DM+RPO, diabetic rats receiving 10% topical red palm oil

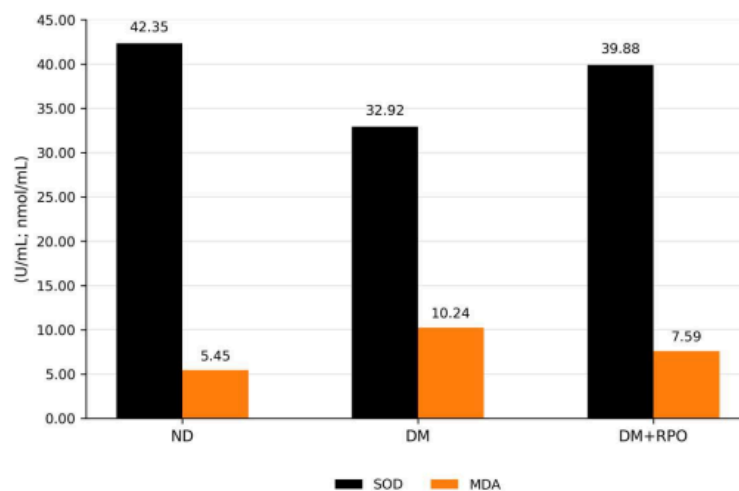


Figure 1. Mean cutaneous SOD activity and MDA concentration by experimental group. ND, non-diabetic control; DM, diabetic control; DM+RPO, diabetic rats receiving 10% topical red palm oil.

The overall distribution of SOD and MDA is illustrated in Figure 1. In diabetic control rats, the reduction in SOD occurred together with an increase in MDA, indicating a shift toward oxidative stress in diabetic skin. In contrast, the RPO-treated group showed a higher mean SOD value and a lower mean MDA value than the diabetic control group, suggesting partial restoration of antioxidant-oxidant balance after topical RPO application.

Assumption testing and global statistical comparisons are summarized in Table 2. SOD data met the assumptions for parametric analysis, with Shapiro-Wilk p values of 0.087, 0.280, and 0.617 in the ND, DM, and DM+RPO groups, respectively, and a non-significant Levene's test ($p=0.053$). One-way ANOVA showed a significant overall difference in SOD activity among groups ($F=23.696$, $p<0.001$). Tukey testing indicated that the diabetic control group differed significantly from both the ND and DM+RPO groups, whereas the ND and DM+RPO groups did not differ significantly.

MDA data did not meet the normality criterion in the ND ($p=0.037$) and DM+RPO ($p=0.005$) groups; therefore, non-parametric analysis was used. The Kruskal-Wallis test demonstrated a significant overall difference in MDA concentration among groups ($H=16.358$, $p<0.001$). Mean ranks were lowest in the ND group (6.67), highest in the DM group (21.78), and intermediate in the DM+RPO group (13.56). Pairwise Mann-Whitney U testing with Holm correction indicated significant differences among all three groups.

Table 2. Assumption testing and global between-group comparison of oxidative-stress biomarkers.

Analysis	SOD activity	MDA concentration
Shapiro-Wilk p (ND / DM / DM+RPO)	0.087 / 0.280 / 0.617	0.037 / 0.316 / 0.005
Levene's test p value	0.053	0.332
Global test	One-way ANOVA	Kruskal-Wallis
Test statistic	$F = 23.696$	$H = 16.358$
Global p value	<0.001	<0.001
Post hoc interpretation	DM differed from ND and DM+RPO; ND and DM+RPO were not significantly different	All pairwise comparisons were significant after Holm correction

ND, non-diabetic control; DM, diabetic control; DM+RPO, diabetic rats receiving 10% topical red palm oil. SOD was followed by Tukey's test. MDA was followed by pairwise Mann-Whitney U tests with Holm correction.

Across all animals, SOD activity was negatively correlated with MDA concentration ($r=-0.544$, $p=0.003$; $n=27$). This inverse correlation indicates that higher antioxidant enzyme activity was associated with lower lipid peroxidation in the combined dataset (Table 3).

Table 3. Correlation between cutaneous SOD activity and MDA concentration ($n=27$).

Statistic	SOD vs MDA
Pearson correlation coefficient (r)	-0.544
p value	0.003
Number of observations	27

DISCUSSION:

This study demonstrated that STZ-induced diabetes was associated with a deteriorated cutaneous oxidative-stress biomarker profile, characterized by lower SOD activity and higher MDA concentration. Topical treatment with 10% RPO improved both markers relative to diabetic control. The increase in SOD suggests enhancement or preservation of antioxidant defense, whereas the reduction in MDA suggests lower lipid peroxidation. Because MDA in the RPO-treated group remained higher than in non-diabetic controls, the effect should be interpreted as partial biochemical improvement rather than complete normalization.

An important feature of the results is the different degree of response between SOD and MDA. SOD activity in the RPO-treated diabetic group approached the non-diabetic control value and was not significantly different from it, suggesting that topical RPO may help preserve or restore enzymatic antioxidant defense. In contrast, MDA decreased significantly after RPO treatment but remained significantly higher than in the non-diabetic group. This pattern suggests that the antioxidant defense system may recover more readily than the downstream lipid-peroxidation burden, or that the diabetic condition continues to generate oxidative damage despite topical antioxidant exposure.

The combined interpretation of SOD and MDA is stronger than either marker alone. If SOD were assessed without MDA, the findings would only suggest improved antioxidant capacity. If MDA were assessed without SOD, the findings would only suggest reduced lipid peroxidation. The concurrent increase in SOD and decrease in MDA provides a more coherent antioxidant-oxidant profile, supporting the interpretation that RPO influenced both the protective and injury-related components of oxidative stress in diabetic skin.

The diabetic control findings are consistent with the known role of oxidative stress in diabetes-associated tissue injury. Hyperglycemia can stimulate mitochondrial ROS production and promote inflammatory and metabolic disturbances that weaken tissue homeostasis.^{3,4} Excess ROS can overwhelm enzymatic antioxidants and damage polyunsaturated fatty acids in cellular membranes, resulting in increased MDA formation.⁵ Thus, the observed pattern of decreased SOD and increased MDA in diabetic skin provides a coherent biochemical signal of redox imbalance in this animal model.

The response to topical RPO is biologically plausible. RPO contains carotenoids, tocopherols, tocotrienols, polyphenols, phytosterols, squalene, and coenzyme Q10, which may contribute to antioxidant protection.^{9,10} Carotenoids and vitamin E isomers can participate in free-radical scavenging and protection of lipid compartments. The topical route is also relevant because the skin barrier contains lipid-rich structures that may interact with lipophilic components of RPO. Although the present study did not identify which RPO constituents penetrated the tissue or mediated the effect, the direction of SOD and MDA changes is compatible with local antioxidant activity.

The diabetic skin environment is not only a passive target of oxidative injury. Skin barrier dysfunction, impaired microcirculation, low-grade inflammation, and delayed cellular turnover may all contribute to local vulnerability. For this reason, a topical preparation that improves local redox balance may be useful as an adjunctive strategy for skin protection, even before a wound develops. Nevertheless, this possible preventive or supportive role should be tested directly in future studies using barrier-function outcomes, transepidermal water loss, moisture measurement, histological thickness, and inflammatory markers.

The findings also align with previous reports of RPO or palm-derived bioactives in oxidative-stress models. In a hepatic injury model, RPO reduced oxidative challenge and inflammatory responses through pathways involving NF-kappaB and Nrf2/GCL/HO-1 signaling.¹¹ Another study reported that RPO pretreatment increased GSH, SOD, and catalase while reducing MDA in Wistar rats.¹² Although those studies were not designed as topical skin experiments, they support the broader concept that RPO can influence antioxidant systems and lipid peroxidation in vivo.

Comparison with topical antioxidant interventions strengthens the interpretation of the present findings. Chrysin and metformin have been reported to reduce oxidative damage and improve antioxidant markers in diabetic wound models.^{6,7} Topical *Phaleria macrocarpa* extract, sinapic acid-loaded gel, glucose-responsive antioxidant hydrogel, ROS-scavenging materials, and multifunctional hydrogels have also been discussed as approaches to modulate oxidative stress in diabetic tissue or wound repair.^{8,17-20} These studies show that local antioxidant modulation is a plausible therapeutic direction, but the present work differs by focusing specifically on RPO and by assessing biochemical markers in diabetic skin rather than wound-closure outcomes.

Palm-derived tocotrienol and vitamin E studies are particularly relevant. Topical palm tocotrienol-rich fraction has been reported to enhance cutaneous wound healing in diabetic mice, and tocotrienol-rich nanoemulsified systems have demonstrated keratinocyte-related and tissue-regeneration effects.^{13,15} Reviews and experimental reports also support the biological relevance of tocotrienols and palm oil fractions in topical wound contexts.^{14,16,21,22} The present study contributes to this literature by using red palm stearin/RPO as the topical material and by presenting SOD-MDA evidence in diabetic skin.

Compared with purified antioxidant compounds, whole RPO or red palm stearin may provide a broader mixture of lipophilic constituents. This can be advantageous because multiple antioxidant components may act through complementary mechanisms; however, it also creates standardization challenges. Without compositional profiling, it is not possible to determine whether the observed effect was driven mainly by carotenoids, tocopherols, tocotrienols, or other minor constituents. This is an important consideration for journal reviewers because natural-product research requires reproducibility of the active material.

From a pharmaceutical perspective, the results support the potential of RPO as a natural-product candidate for topical development. However, further formulation work is essential before translation. The present study did not characterize the active-compound content of RPO, the composition of the base cream, the amount of formulation deposited per application, skin penetration, physical stability, irritation potential, or release profile. These factors are

important because topical efficacy depends not only on the intrinsic activity of the bioactive material but also on vehicle behavior, contact time, dermal compatibility, and reproducibility of dosing.

The use of red palm stearin is also relevant from a formulation standpoint. A stearin fraction may influence spreadability, occlusiveness, and residence time on the skin compared with liquid oils. These properties may affect contact between bioactive lipophilic constituents and the stratum corneum. However, because the present work did not test rheology, release, permeation, or stability, the current results should be considered proof-of-concept biochemical data rather than evidence of an optimized topical product.

The study also has value in the context of natural-product development. RPO is an accessible biological resource in tropical countries and contains antioxidant constituents that are relevant to skin protection. Nevertheless, natural-product preparations require standardization to ensure batch-to-batch consistency. Future work should quantify carotenoid, tocopherol, and tocotrienol contents; compare red palm oil with red palm stearin; and determine whether the observed SOD-MDA changes are retained after formulation into a stable topical dosage form.

The negative SOD-MDA correlation supports the expected inverse relationship between antioxidant defense and lipid peroxidation. However, the correlation was calculated across the combined dataset of all groups and should not be interpreted as proof of a direct causal pathway. Group differences may partly drive the association. Future analyses using raw individual-level data could examine within-group correlations and compare Pearson with Spearman coefficients, particularly because MDA did not meet normality assumptions in all groups.

Several limitations should be emphasized. First, this was a small preclinical study using a chemically induced diabetes model, and the findings may not represent the complexity of chronic human diabetic skin disease. Second, the study assessed only SOD and MDA; it did not include histopathology, inflammatory cytokines, collagen deposition, angiogenesis, oxidative DNA damage, or protein oxidation. Third, no wound model was created, and no wound closure or tensile-strength outcome was measured. Therefore, the results **should not be interpreted as direct evidence of wound-healing efficacy**. Fourth, the RPO formulation was not chemically standardized or evaluated for stability and dermal penetration. Future studies should include dose-response evaluation, standardized formulation characterization, irritation testing, histological endpoints, inflammatory and angiogenic markers, and controlled diabetic wound models before clinical translation is considered.

Another important point is that the model used in this study represents diabetic skin under experimental conditions, not a clinical diabetic ulcer. Therefore, the findings should guide future preclinical formulation work rather than immediate clinical recommendations. A stronger next study would include a standardized diabetic wound model, serial wound-area measurement, histopathological assessment, collagen organization, inflammatory markers such as TNF- α or IL-6, angiogenic markers such as VEGF, and oxidative pathway markers such as Nrf2 or HO-1. These additional endpoints would help determine whether the biochemical antioxidant effect observed here translates into structural or functional skin repair.

CONCLUSION:

Topical 10% RPO increased cutaneous SOD activity and reduced MDA concentration in STZ-induced diabetic Wistar rats. These findings indicate that RPO can improve the antioxidant-oxidant biomarker profile in diabetic skin. The study supports further investigation of RPO as a natural-product-based topical antioxidant candidate, but it does not establish clinical benefit or wound-healing efficacy. Additional formulation, safety, histological, inflammatory, mechanistic, and translational studies are required.

CONFLICT OF INTEREST:

The authors declare no conflict of interest regarding this study.

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